

Sam Gunn

CONTACT	Email: gunn@mit.edu Website: https://samuel-gunn.github.io/
POSITIONS	Massachusetts Institute of Technology C.L.E. Moore Instructor of Mathematics (Summer 2026 to Present)
EDUCATION	The University of California, Berkeley Ph.D. in Computer Science (Fall 2020 to Summer 2026) Advisor: Umesh Vazirani The University of Texas at Austin B.S. in Mathematics (Fall 2016 to Spring 2020) Advisors: Scott Aaronson and Cameron Gordon
RESEARCH	1. How to sketch a learning algorithm. <i>In submission.</i> arXiv. 2. Improved pseudorandom codes from permuted puzzles. With Miranda Christ, Noah Golowich, Ankur Moitra, and Daniel Wichs. <i>In submission.</i> ePrint. 3. Black-box crypto is useless for pseudorandom codes. With Sanjam Garg and Mingyuan Wang. <i>TCC 2025 (outstanding paper award).</i> ePrint. 4. An undetectable watermark for generative image models. With Xuandong Zhao and Dawn Song. <i>ICLR 2025.</i> arXiv. 5. SoK: Watermarking for AI-generated content. With Xuandong Zhao et al. <i>S&P 2025.</i> arXiv. 6. Ideal pseudorandom codes. With Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, and Yevgeniy Dodis. <i>STOC 2025 (invited to the SICOMP Special Issue).</i> ePrint. 7. Quantum one-time protection of any randomized algorithm. With Ramis Movassagh. <i>CRYPTO 2025.</i> ePrint. 8. Classical commitments to quantum states. With Yael Kalai, Anand Natarajan, and Agi Villanyi. <i>QIP 2025, STOC 2025.</i> ePrint. 9. Pseudorandom error-correcting codes. With Miranda Christ. <i>CRYPTO 2024.</i> ePrint. 10. Undetectable watermarks for language models. With Miranda Christ and Or Zamir. <i>COLT 2024, RWC 2024.</i> ePrint. 11. How to use quantum indistinguishability obfuscation. With Andrea Coladangelo. <i>STOC 2024.</i> ePrint. 12. Approaching the quantum Singleton bound with approximate error correction. With Thiago Bergamaschi and Louis Golowich. <i>QIP 2024, STOC 2024.</i> arXiv. 13. Commitments to quantum states. With Nathan Ju, Fermi Ma, and Mark Zhandry. <i>QIP 2023, STOC 2023 (invited to the SICOMP Special Issue).</i> ePrint.
AWARDS	• Google PhD Fellowship 2022–2025 • Lotfi A. Zadeh Prize 2024–2025
TALKS	1. How to sketch a learning algorithm • MIT Crypto+AI seminar (May 2026). • Institute for Advanced Study (May 2026). • Flatiron Institute (June 2026). • STOC Theory for Trustworthy and Interpretable AI Workshop (June 2026). • Invited talk at ITC (August 2026).

2. **On undetectable backdoors** at the UT Austin Theory and Alignment group (November 2025).
3. **Pseudorandom error-correcting codes**
 - NTT Research (March 2024).
 - Google Research NYC (April 2024).
 - CRYPTO 2024 (August 2024).
 - NYU (March 2025).
 - Harvard Theory Seminar (September 2025).
 - TCC 2025 (December 2025).
4. **Undetectable watermarks for language models**
 - Google Research Mountain View (July 2023).
 - MIT Cryptography Group (August 2023).
 - NYU Cryptography Seminar (September 2023). Video [here](#).
 - Berkeley Machine Learning Theory Seminar (September 2023).
5. **Quantum one-time protection of any randomized algorithm** at CRYPTO 2025 (August 2025).
6. **Quantum indistinguishability obfuscation**
 - CIQC Spark Talk, March 2024.
 - The University of Ottawa, April 2024.
7. **Approaching the quantum Singleton bound with approximate error correction** at the Simons Quantum Colloquium (December 2022). Video [here](#).
8. **Commitments to quantum states** at STOC 2023 (June 2023). Video [here](#).
9. **Unclonable cryptography** guest lecture for John Wright's graduate quantum complexity course at UC Berkeley (December 2022).
10. **Certified entropy from random circuit sampling** at Berkeley Theory Lunch (March 2022).
11. **On the classical hardness of spoofing linear cross-entropy benchmarking** plenary talk at SQuInT 2020 (February 2020).